Maths Snapshots - Learning Phase 2 (Y4-6)	Name:	Year: 1 2 3 4 5 6 7 8

Notes for teacher

This is a written assessment tool that is aligned with the new NZ maths curriculum, created by Marie Hirst and Dr Jo Knox,

It is a 'snapshot' designed to quickly formatively assess what a student can do in some threads of the number strand in phase 2 (but not any other strands).

It is a low-stakes tool to help provide opportunities for observation and identify next teaching steps. It is not a high-stakes assessment for making summative judgements about math and it is not standardised or norm referenced.

DO NOT complete the assessment all at once. It is not intended to be completed in one go, but to dip into the maths topics when appropriate. Please print off each topic focus sheet as and when it is needed.

N.B. To assess the recall of multiplication and division facts in section 4, please also use the additional mult/div basic facts slideshow. Each question is displayed for 4 seconds. If the students cannot recall the fact, please ask them to miss that question and move on to the next.

Section	Sub-strand	Topic
1	Number Structure	Place value: read and write numbers (and say number 1, 10, ,100 before and after)
2	Number Operations	Round numbers
3	Number Operations	Add and subtract whole numbers
4	Number Operations	Recall multiplication and division facts
5	Number Operations	Multiply and divide whole numbers
6a	Rational Numbers	Compare and order decimals
6b	Rational Numbers	Identify and convert fractions, decimals and percentages
7	Rational Numbers	Multiply and divide numbers by 10 and 100 to make decimals and whole numbers
8	Rational Numbers	Add and subtract decimals
9	Rational Numbers	Convert between mixed numbers and improper fractions
10	Rational Numbers	Find a fraction of an amount
11	Rational Numbers	Compare and order fractions, identify equivalence and represent in simplest form
12	Rational Numbers	Add and subtract fractions

Maths Snapshots - Learning Phase 2 (Y4-6)	Name:	Year: 3 4 5 6 7 8

1. Place Value: Identify and write what number comes 1, 10 or 100 before it (less than) and after it (more than).

	Y3 Numbers to 1,000		Y4 Numbers to 10,000		Y5 Numbers to 100,000		Y6 Numbers to 1,000,000	
	870	**409**	**4,732**	**2,099**	**73,481**	**81,097**	**628,547**	**789,099**
1 more than	871	410	4,733	2,100	73,482	81,098	628,548	789,100
1 less than	869	408	4,731	2,098	73,480	81,096	628,546	789,098
10 more than	880	419	4,742	2,109	73,491	81,107	628,557	789,109
10 less than	860	399	4,722	2,089	73,471	81,087	628,537	789,089
100 more than	970	509	4,832	2,199	73,581	81, 197	628,647	789,199
100 less than	770	309	4,632	1,999	73,381	80,997	628,447	788,999

Expand this number	759 = 700 + 50 + 9	5,624 = 5000 + 600 + 20 + 4	27,431 = 20,000 +7000 + 400 + 30 + 1	729,976 = 700,000 + 20,000 + 9000 + 900 + 70 + 6
How many **tens** are in **<u>all of</u>** the number?	759 = 75	5,624 = 562	27,431 = 2743	729,976 = 72997
How many **hundreds** are in **<u>all of</u>** the number?	759 = 7	5,624 = 56	27,431 = 274	729,976 = 7299

Maths Snapshots - Learning Phase 2 (Y4-6)	Name:	Year: 3 4 5 6 7 8

2. Round whole numbers and decimals

	Y3 Numbers to 1,000	Y4 Numbers to 10,000	Y5 Numbers to 100,000	Y6 Numbers to 1,000,000
Number	759	5,624	27,431	729,976
Round the number to the **nearest ten**	760	5,620	27,430	729,980
Round the number to the **nearest hundred**	800	5,600	27,400	730,000
Round the number to the **nearest thousand**		6,000	27,000	730,000
Round the number to the **nearest ten thousand**			30,000	730,000

			Decimal tenths	Tenths and hundredths
Number			6.4	13.53
Round to the nearest whole number			6	14
Round to the nearest. Tenth (1dp)				13.5

Maths Snapshots - Learning Phase 2 (Y4-6)	Name:	Year: 3 4 5 6 7 8

3. Add and Subtract Whole Numbers

	Y3 two-digit numbers (with renaming)	Y4 three-digit numbers (no renaming)	Y5 numbers up to 10,000	Y6 any whole numbers
Addition	37 + 18 = 55	325 + 124 = 449	4,125 + 377 = 4,502	26,578 + 13,425 = 40,003
		Y4 three-digit numbers (renaming)		
		484 + 347 = 831		
Subtraction	Y3 two-digit numbers (with renaming)	Y4 three-digit numbers (no renaming)	8,037 – 1,648 = 6,389	242,527 – 58,382 = 184,145
	43 - 28 = 15	784 – 532 = 252		
		Y4 three-digit numbers (renaming)		
		623 - 258 = 365		

Maths Snapshots - Learning Phase 2 (Y4-6)	Name:	Year: 3 4 5 6 7 8

4. Recall Multiplication and Division Facts.

	Y3 recall x ÷ facts for 2s, 3s, 5s and 10s	Y4 recall x ÷ facts for 4s and 6s	Y5 recall x ÷ facts for 7s, 8s, 9s	Y6 recall all x ÷ facts to 10 x 10
4a. Multiplication facts	1) 2 x 8 = 16 2) 3 x 6 = 18 3) 6 x 5 = 30 4) 7 x 10 = 70 5) 5 x [8] = 40	1) 4 x 6 = 24 2) 4 x 3 = 12 3) 8 x 4= 32 4) 3 x 6 = 18 5) 6 x [7] = 42	1) 7 x 7 = 49 2) 6 x 8 = 48 3) 9 x 8 = 72 4) 7 x 9 = 63 5) 8 x [7] = 56	1) 3 x 7 = 21 2) 7 x 8 = 56 3) 9 x 6 = 54 4) 4 x 9 = 36 5) 8 x [8] = 64
4b. Division facts	6) 14 ÷ 2 = 7 7) 40 ÷ 5 = 8 8) 60 ÷ 10 = 6 9) 24 ÷ 3 = 8 10) 20 ÷ [4] = 5	6) 24 ÷ 6 = 4 7) 12 ÷ 4 = 3 8) 42 ÷ 6= 7 9) 36 ÷ 4 = 9 10) 16 ÷ [4] = 4	6) 28 ÷ 7 = 4 7) 56 ÷ 8 = 7 8) 64 ÷ 8 = 8 9) 63 ÷ 9 = 7 10) 72 ÷ [9] = 8	6) 49 ÷ 7 = 7 7) 32 ÷ 8 = 4 8) 27 ÷ 3 = 9 9) 54 ÷ 9 = 6 10) 63 ÷ [7] = 9

Maths Snapshots - Learning Phase 2 (Y4-6)	Name:	Year: 3 4 5 6 7 8

5a. Multiplication Operations

Y3 1-digit x 1-digit	Y4 1-digit x 1-digit	Y5 3-digit x 1-digit	Y6 multi-digit numbers
4 x 6 = 24	7 x 8 = 56	453 x 6 = 2,718	54 x 132 = 7,128
Y3 simple 1-digit x 2-digit	**Y4 2-digit x 1-digit**	**Y5 2 x 2-digit numbers**	
2 x 23 = 46	54 x 3 = 162	26 x 45 = 1,170	

5b. Division Operations

Y3 Divide by a 1-digit divisor (no reminders)	Y4 Divide up to a 3-digit number by a 1-digit divisor (no remainders)	Y5 Divide up to a 3-digit number by a 1-digit divisor (with remainders)	Y6 Divide up to a 4-digit number by a 1-digit divisor (with remainders)
24 ÷ 3 = 8	65 ÷ 5 = 13	79 ÷ 3 = 26 r1 or 26.33333333	293 ÷ 7 = 41 r6 or 41.857
32 ÷ 4 = 8	252 ÷ 3 = 84	194 ÷ 6 = 32 r2 or 32.33333333	4,154 ÷ 8 = 519 r2 or 519.25

Maths Snapshots - Learning Phase 2 (Y4-6)	Name:	Year: 3 4 5 6 7 8

6a. Compare and Order Decimals: For each pair of numbers, circle the **number which is the largest.**

Y4 Decimal tenths (1dp)	Y5 Decimal tenths and hundredths	Y6 Decimals to 2 decimal places (2dp)
0.6 or 0.40	0.7 or 0.32	0.8 or 0.57
0.2 or 0.31	0.27 or 0.395	0.56 or 0.811
0 or 0.5	0 or 0.05	0 or 0.07
0.2 or 0.4	0.3 or 0.10	0.4 or 0.6

6b Decimals, Fractions and Percentage Conversions: Complete the table below

fraction	decimal	percentage
example $^{72}/_{100}$ ⟷	**0.72**	⟷ *72%*

Y4 tenths			Y5 tenths and hundredths			Y6 fractions, decimals & percentages (2d.p.)		
fraction	**decimal**		**fraction**	**decimal**		**fraction**	**decimal**	**percentage**
7/10	**0.7**		54/100	**0.54**		25/100 or 1/4	**0.25**	25%
$^{2}/_{10}$	0.2		$^{43}/_{100}$	0.43		$^{6}/_{10}$	0.6	60%
8/10	**0.8**		3/100	**0.03**		85/100	0.85	**85%**

Maths Snapshots - Learning Phase 2 (Y4-6)	Name:	Year: 3 4 5 6 7 8

7. Multiply and divide numbers by 10 and 100 to make decimals

Y4 Divide whole numbers by 10 to make decimals	Y5 Divide whole numbers by 10 & 100 to make decimals	Y6 Divide numbers by 10 and 100 to make decimals
23 ÷ 10 = 2.3	56 ÷ 10 = 5.6	35.4 ÷ 10 = 3.54
137 ÷ 10 = 13.7	328 ÷ 100 = 3.28	75 ÷ 100 = 0.75
		Y6 Multiply decimals by 10 & 100 to make whole numbers
		0.58 x 10 = 5.8
		6.73 x 100 = 673

Maths Snapshots - Learning Phase 2 (Y4-6)	Name:	Year: 3 4 5 6 7 8

8. Add and Subtract Decimals

	Y4 decimals tenths (1dp) (no renaming)	Y5 whole numbers and decimals to 2dp	Y6 whole numbers and decimals to 2dp
Adding decimals	1.6 + 0.2 = 1.8	2.42 + 1.5 = 3.92	215.7 + 43.59 = 259.29
	Y4 decimal tenths (1dp) (with renaming)		
	0.8 + 0.4 = 1.2		
	Y4 Decimals tenths (1dp) (no renaming)	**Y5 whole numbers and decimals to 2dp**	**Y6 whole numbers and decimals to 2dp**
Subtracting decimals	0.8 – 0.3 = 0.5	16.55 - 2.4 = 14.15	42.25 - 21.83 = 20.42
	Y4 decimal tenths (1dp) (with renaming)		
	1.3 – 0.6 = 0.7		

Maths Snapshots - Learning Phase 1 (Y4-6)	Name:	Year: 3 4 5 6 7 8

9. Rational Numbers: Convert between mixed numbers and improper fractions

	Y4 with denominators 2,3,4,5,6 &10	Y5 with denominators up to 10	Y6 with any denominators
Write these improper fractions as mixed numbers	$^{11}/_{3}$ = $3^{2}/_{3}$	$^{23}/_{8}$ = $2^{7}/_{8}$	$^{34}/_{11}$ = $3^{1}/_{11}$
Write these mixed numbers as improper fractions	4 $^{1}/_{5}$ = $^{21}/_{5}$	2 $^{6}/_{7}$ = $^{20}/_{7}$	4 $^{3}/_{20}$ = $^{83}/_{20}$

10. Rational Numbers: Find a fraction (or percentage) of a number

Y4 Find a unit fraction of a number using mult/div facts	Y5 Find a fraction of a number using mult/div facts	Y6 Find a fraction or a percentage of a number
$^{1}/_{5}$ of 45 = 9	$^{3}/_{5}$ of 35 = 21	$^{3}/_{8}$ of 72 = 27
$^{1}/_{3}$ of 24 = 8	$^{5}/_{6}$ of 30 = 25	$^{3}/_{5}$ of $\boxed{30}$ = 18
¼ of $\boxed{12}$ = 3	$^{2}/_{3}$ of $\boxed{15}$ = 10	30% of 220 = 66

Maths Snapshots - Learning Phase 1 (Y4-6)	Name:	Year: 3 4 5 6 7 8

11. Rational Numbers: Compare and order fractions, identify equivalence and represent in simplest form

Y4 related denominators of 2,4 & 8, 3 & 6 or 5 &10	Y5 denominators of 2,3,4,5,6,8,10,12 or 100	Y6 (Y5 **and** represent fractions in their simplest form)
Circle the **largest** fraction (or if both fractions are equivalent put a **=** between the two fractions)		Write these fractions in their simplest form
$^{2}/_{4}$ $^{5}/_{8}$	$^{2}/_{4}$ $^{5}/_{12}$	$^{6}/_{8}$ = $^{3}/_{4}$
$^{2}/_{3}$ = $^{4}/_{6}$	$^{6}/_{10}$ $^{42}/_{100}$	$^{8}/_{12}$ = $^{2}/_{3}$
$^{2}/_{5}$ $^{3}/_{10}$	$^{2}/_{6}$ = $^{4}/_{12}$	$^{80}/_{100}$ = $^{4}/_{5}$

12. Rational Numbers: Add and Subtract Fractions

Y3 with unit fractions	Y4 with same denominators to make up to 1 whole	Y5 with same denominators to make more than 1 whole	Y6 with same or related denominators
$^{1}/_{8} + ^{1}/_{8} = ^{2}/_{8}$	$^{2}/_{4} + ^{2}/_{4} = ^{4}/_{4}$ or 1	$^{2}/_{3} + ^{2}/_{3} = ^{4}/_{3}$ or $1^{1}/_{3}$	$^{2}/_{3} + ^{2}/_{6} = ^{6}/_{6}$ or 1
$^{1}/_{3} + ^{1}/_{3} + ^{1}/_{3} = ^{3}/_{3}$ or 1	$^{1}/_{6} + ^{2}/_{6} + ^{2}/_{6} = ^{5}/_{6}$	$^{5}/_{6}+^{2}/_{6}+^{2}/_{6} = ^{9}/_{6}$ or $1^{3}/_{6}$ Or 1½	$^{3}/_{8} + ¼ = ^{5}/_{8}$
$^{1}/_{6}+^{1}/_{6}+^{1}/_{6}+^{1}/_{6} = ^{4}/_{6}$	$^{7}/_{8} - ^{3}/_{8} = ^{4}/_{8}$ or ½	$1\,^{3}/_{8} - ^{4}/_{8} = ^{7}/_{8}$	$^{4}/_{5} - ^{3}/_{10} = ^{5}/_{10}$ or ½